

Physics Midterm

3/16/25

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Unit 0: $q = 1.16 \rightarrow 10^{-5} \text{ C}$ $r = 8 \text{ cm} \rightarrow .08 \text{ m}$ $k = 8.99 \cdot 10^9$
 $E = \frac{k|q|}{r^2}$ $0 = \frac{(8.99 \cdot 10^9)(1.16 \cdot 10^{-5})}{.08^2}$
 $= 2.81 \cdot 10^6 \text{ N/C}$
 $E = (1.16 \cdot 10^{-5}) \cdot 2.81 \cdot 10^6 \text{ N/C} \rightarrow 2.81 \text{ N}$

b) $-2q$ at $x = 0 \text{ cm}$
 $+q$ at $x = 5 \text{ cm}$ $E_{\text{net}} = -5.70 \times 10^3 \text{ N/C}$
 $+3q$ at $x = 8 \text{ cm}$
 $-q$ at $x = 10 \text{ cm}$

2) a. $q_a = \ominus$ $q_b = \oplus \rightarrow$ repulsive (out)
 $q_b = \oplus$ $q_c = \oplus \rightarrow$ repulsive (out)
 $q_c = \ominus$ $q_d = \ominus \rightarrow$ attractive (in)
 $q_d = \ominus$ $q_e = \ominus \rightarrow$ attractive (in)

b) net force = 0 N due to symmetry, force from all 4 wires cancel each other out, net

3) $q = 1.6 \times 10^{-19} \text{ C}$
 $V = 40 \text{ kV}$ $v = \sqrt{\frac{2qV}{m_e}}$ $V = \sqrt{\frac{2 \cdot 1.60 \cdot 10^{-19} \cdot 40 \cdot 10^3}{9.109 \cdot 10^{-31}}}$
 $m_e = 9.11 \times 10^{-31}$ $v = 1.14 \times 10^8 \text{ m/s}$

$$4) a) \Delta V = E \cdot d$$

$$E = 7.5 \times 10^4 \text{ V/m}$$

$$d = 4 \text{ cm} \rightarrow 0.04 \text{ m}$$

$$\Delta V = (7.5 \times 10^4) \cdot (0.04)$$

$$\Delta V = 3000 \text{ V}$$

$$b) V = E \cdot d$$

$$E = 7.5 \times 10^4 \text{ V/m} \quad x = 1 \text{ cm} = 0.01 \text{ m}$$

$$V = (7.5 \times 10^4) \cdot (0.01)$$

$$V = 750 \text{ V}$$

$$c) \frac{\Delta V}{d} = E$$

$$V = 80 \cdot 10^{-3} \text{ V}$$

$$d = 1.40 \cdot 10^{-4} \text{ m}$$

$$E = \frac{80 \cdot 10^{-3}}{1.40 \cdot 10^{-4}} = 5.71 \cdot 10^6 \text{ V/m}$$

$$5) E = \frac{V}{d}$$

$$q = 2e$$

$$e = 1.602 \times 10^{-19} \text{ C}$$

$$32 \times 10^3 \text{ eV} \times (1.602 \times 10^{-19})$$

$$V = 5.126 \times 10^{-15} \text{ J}$$

$$E = \frac{V}{d} = \frac{10 \times 10^3}{0.02}$$

$$= 5 \times 10^5 \text{ V/m}$$

$$V = 1.6 \times 10^4 \text{ V} = 16 \text{ kV}$$

$$6) C_{\min} = \frac{(8.44 \times 10^9) (2 \times 1.602 \times 10^{-19}) (79 \times 1.602 \times 10^{-19})}{8.01 \times 10^{-18}}$$

$$\hookrightarrow \frac{q_c (2e) (79e)}{E} \uparrow$$

$$= 4.55 \times 10^{-14} \text{ or } 45.5 \text{ fF}$$

$$\text{Unit 1: } C = \frac{Q}{V}$$

$$Q = \frac{3 \times 10^{-6} \text{ C}}{120 \text{ V}}$$

$$1)$$

$$= 2.5 \times 10^{-8}$$

$$C = 25.0 \text{ nF}$$

$$2) E = \frac{1}{2} CV^2$$

$$a)$$

$$E = \frac{1}{2} (10 \times 10^{-6} \text{ F}) \times (4 \times 10^3 \text{ V})^2$$

$$E = 4.05 \times 10^2 \text{ J}$$

$$E = 405 \text{ J}$$

$$b) E = \frac{1}{2} CV^2$$

$$V = \sqrt{\frac{2E}{C}} \rightarrow V = \sqrt{\frac{2 \times 40 \text{ J}}{8 \times 10^{-6} \text{ F}}}$$

$$V = \sqrt{10^7}$$

$$V = 3162 \text{ V}$$

$$3) E = \frac{1}{2} C_{\text{total}} V^2$$

$$C_{\text{total}} = \frac{2E}{V^2}$$

$$C_{\text{individual}} = \frac{0.988}{4}$$

$$= 0.247$$

$$C_{\text{total}} = \frac{2 \times 40}{(9.0 \times 10^3)^2}$$

$$= 9.88 \times 10^{-7} \text{ F} = \frac{0.988}{0.988} \text{ pF}$$



$$Q_{\text{individual}} = 0.247 \times 10^{-6} \text{ F} \times 9 \times 10^3 \text{ V}$$

$$= 2.22 \times 10^{-3} = 2.22 \text{ mC}$$

0.247 μF = required capacitance on each capacitor

2.22 mC = charged stored on each capacitor

The designer would prefer to connect the capacitors in parallel to maintain a higher total capacitance, store more charge and prevent excessive voltage across each capacitor

$$4. 4) I = \frac{V}{R}$$

$$R_{\text{total}} = 1000 \Omega + 3 \Omega = 1003 \Omega$$

$$I = \frac{3 \text{ V}}{1003 \Omega} = 2.99 \text{ mA}$$

$$5) P = I^2 \times R = (2.99 \text{ mA})^2 \times 3 \Omega = 26.82 \text{ mW}$$

Unit 2

$$1) a) \frac{1}{R_{\text{tot}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$\downarrow \quad \quad \downarrow \quad \quad \downarrow$
 $2 \text{ k}\Omega \quad 10 \text{ k}\Omega \quad 5 \text{ k}\Omega$

$$\frac{1}{R_3} = \frac{1}{2000} - \left(\frac{1}{10000} + \frac{1}{5000} \right)$$

$$R_3 = \frac{1}{0.0002} = 5000 \Omega = 5 \text{ k}\Omega$$

$$b) I_{\text{tot}} = \frac{V}{R_{\text{tot}}} \quad V = 12 \text{ V}$$

$$I_{\text{tot}} = \frac{12 \text{ V}}{2000 \Omega}$$

$$I_{\text{tot}} = 0.006 \text{ A} = 6 \text{ mA}$$

$$\begin{aligned} c) \quad I_1 &= \frac{12 \text{ V}}{R_1} = \frac{12 \text{ V}}{10 \text{ k}\Omega} = \frac{12}{10000} = 1.2 \text{ mA} \\ I_2 &= \frac{12 \text{ V}}{R_2} = \frac{12 \text{ V}}{5 \text{ k}\Omega} = \frac{12}{5000} = 2.4 \text{ mA} \\ I_3 &= \frac{12 \text{ V}}{R_3} = \frac{12 \text{ V}}{5 \text{ k}\Omega} = \frac{12}{5000} = 2.4 \text{ mA} \end{aligned} \quad \left. \vphantom{\begin{aligned} I_1 \\ I_2 \\ I_3 \end{aligned}} \right\} 6 \text{ mA total current}$$

2) a) $V_{\text{total}} = V_1 + V_2 = 1.5V + 1.5V = 3V$

Ohm's law $R = 0.5k\Omega = 500\Omega$

$I = \frac{V}{R} = \frac{3V}{500\Omega} = 0.006A = 6mA$

power calculation $P = V \times I = 3V \times 6mA = 18mW$

Series circuit and parallel circuit have same current (6mA) and power 18mW

b) $R_{\text{total}} = 0.5\Omega + 0.5\Omega = 1\Omega$

$R_{\text{total}} = 1\Omega + 500\Omega = 501\Omega$

$V_{\text{total}} = 1.5V + 1.5V = 3V$

$I = \frac{3V}{501\Omega} = 5.98mA$

$P = V_R \times I \quad V_R = (5.98)(500\Omega) = 2.99V$

$P = 2.99V \times 5.98mA \approx 17.9mW$

$R_{\text{branch}} = 500.5\Omega$

$R_{\text{total}} = \frac{500.5}{2} = 250.25\Omega$

$I_{\text{total}} = \frac{3V}{250.25} \approx 11.99mA$

$I_1 = I_2 = \frac{11.99}{2} = 5.99mA$

$P = V_R \times I_{\text{total}}$

$P = 3V \times 11.99mA \approx 35.97mW$

parallel circuit =
current = 11.99mA
power = 35.97mW

Series circuit =
current = 5.98mA
power = 17.9mW

2. a) $V_{\text{total}} = (3 \times 1.58 \text{ V}) + 1.53 \text{ V}$
 $= 6.27 \text{ V}$

$R_{\text{total}} = 0.16 \Omega + 10 \Omega = 10.16 \Omega$

$I = \frac{6.27 \text{ V}}{10.16 \Omega} = 0.6175 \text{ A}$

b) $P = V_{\text{load}} \times I$

$V_{\text{load}} = (0.618 \text{ A})(10 \Omega) = 6.18 \text{ V}$

$P_{\text{load}} = 6.18 \text{ V} \times 0.619 \text{ A} = 3.82 \text{ W}$

c) $V_{\text{load}} = \sqrt{5} = 2.236 \text{ V}$

$I_{\text{new}} = \frac{2.236 \text{ V}}{10 \Omega} = 0.224 \text{ A}$

$R_{\text{total, new}} = \frac{6.27 \text{ V}}{0.224 \text{ A}} = 28.04 \Omega$

$R_{\text{internal, new}} = 28.04 \Omega - 10 \Omega = 18.04 \Omega$

When bad cell goes away $R_{\text{bad cell}} = 18.04 \Omega - 0.06 \Omega = 17.98 \Omega$

Unit 3:

1)

Case	V	A	F
a	↓	↑	0
b	↓	↑	0
c	↑	↓	↑

$$2) \quad F = qvB \sin(\theta) \rightarrow \frac{F}{qvB} = \sin \theta$$

$$\sin \theta = \frac{1.2 \times 10^{-16}}{(1.6 \times 10^{-19})(4 \times 10^3)(1.25)}$$

$$\theta = \sin^{-1}(1.25) \rightarrow 16.1 + 180 - 16.1$$

$$= 10.1 + 169.9 = 169.9$$

$$\theta = 10.1^\circ \text{ or } 169.9^\circ$$

$$3) \quad q = \frac{(2.66 \times 10^{-26})(5 \times 10^6)}{(0.231)(1.20)} \quad \left| \begin{array}{l} \text{elementary charge} \\ e = 1.60 \times 10^{-19} \text{ C} \end{array} \right.$$

$$q = 4.8 \times 10^{-19} \text{ C}$$

$$\frac{q}{e} = \frac{4.8 \times 10^{-19} \text{ C}}{1.6 \times 10^{-19} \text{ C}} = 3$$

$$r = \frac{mv}{qB} \quad m \frac{18}{16} = 1.125$$

$$r_{18} = r_{16} \times \frac{m_{18}}{m_{16}}$$

$$r_{18} = 0.231 \times 1.125$$

$$\approx 0.260 \text{ m}$$